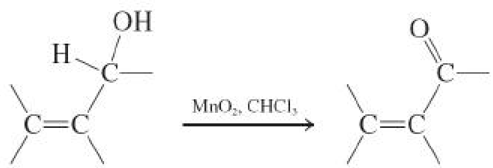
New Reactions

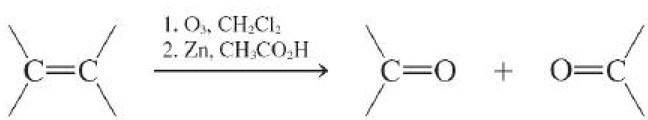
Synthesis of Aldehydes and Ketones

1. Oxidation of Alcohols (Section 17-4)

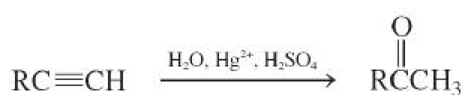
Allylic oxidation



2. Ozonolysis of Alkenes (Section 17-4)



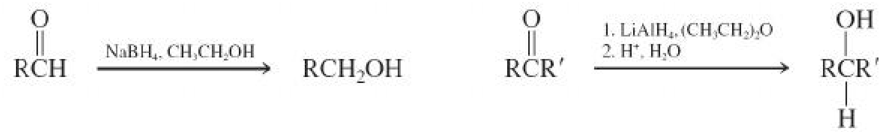
3. Hydration of Alkynes (Section 17-4)



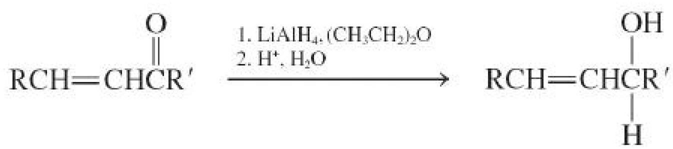
4. Friedel-Crafts Acylation (Section 17-4)

Reactions of Aldehydes and Ketones

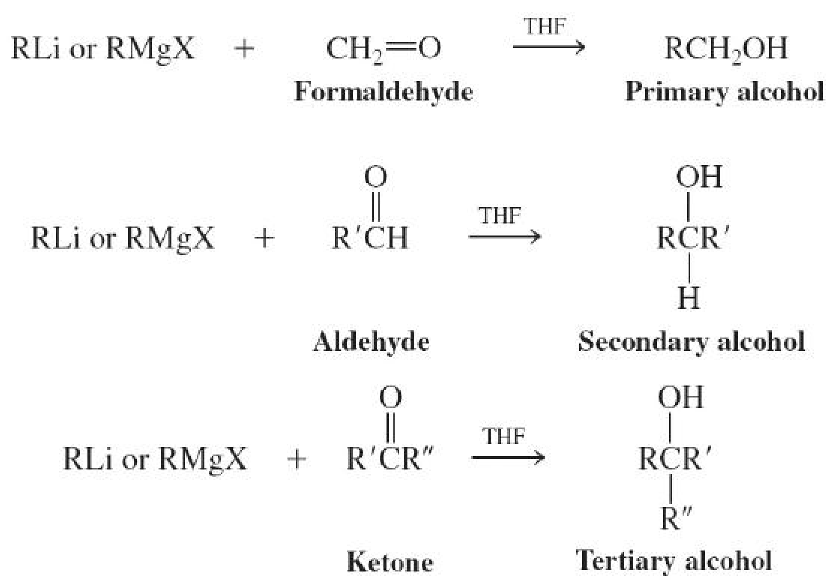
5. Reduction by Hydrides ([Section 17-5](#))



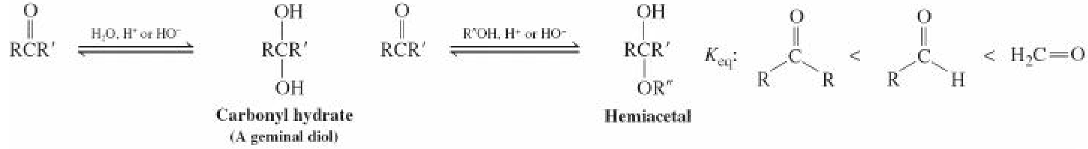
Selectivity



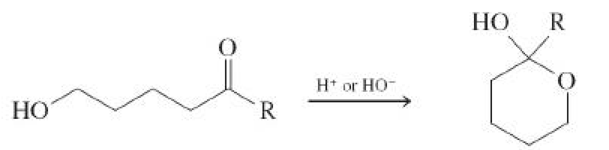
6. Addition of Organometallic Compounds ([Section 17-5](#))



7. Addition of Water and Alcohols—Hemiacetals ([Sections 17-6](#) and [17-7](#))

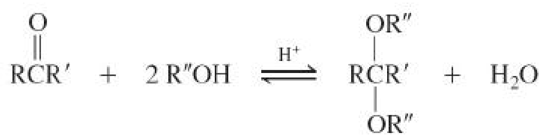


Intramolecular addition



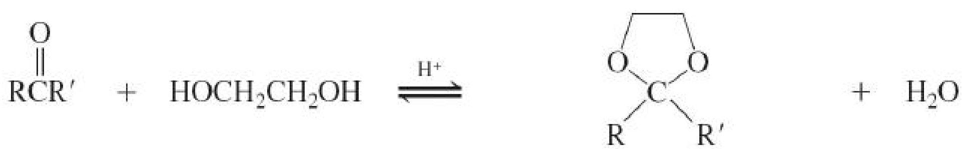
Cyclic hemiacetal

8. Acid-Catalyzed Addition of Alcohols—Acetals (Sections 17-7 and 17-8)



Acetal

Cyclic acetals

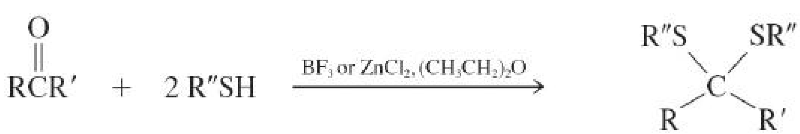


Ketone

Ketone, protected as cyclic acetal (Stable to base, $LiAlH_4$, RMgX)

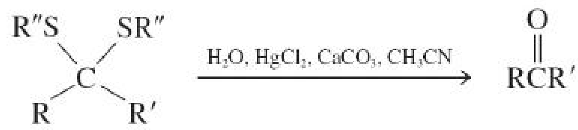
9. Thioacetals (Section 17-8)

Formation



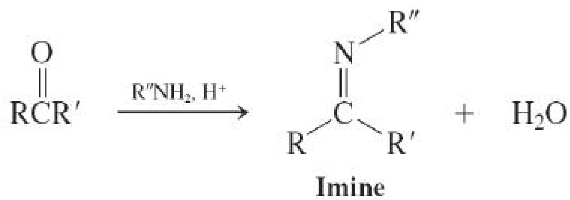
(Stable to aqueous acid, base, $LiAlH_4$, RMgX)

Hydrolysis

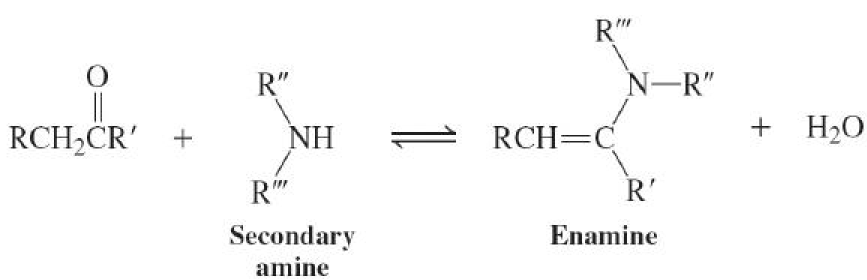


10. Raney Nickel Desulfurization (Section 17-8)

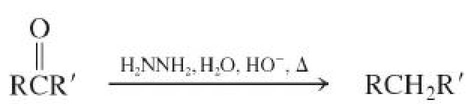
11. Addition of Amine Derivatives (Section 17-9)



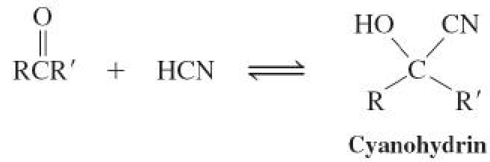
12. Enamines (Section 17-9)



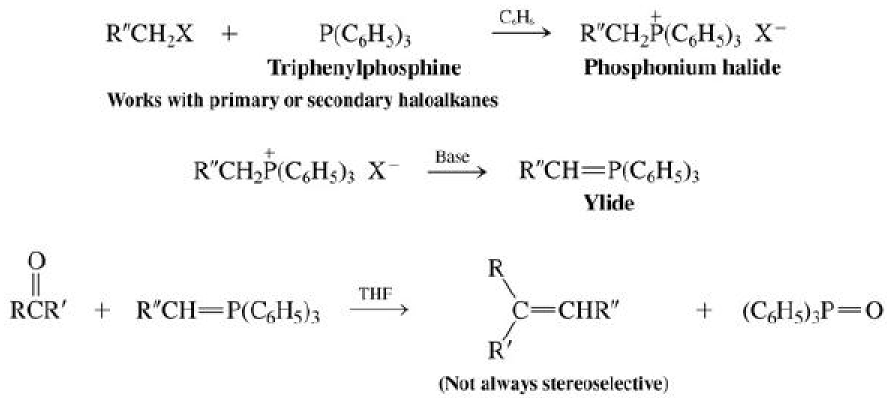
13. Wolff-Kishner Reduction (Section 17-10)



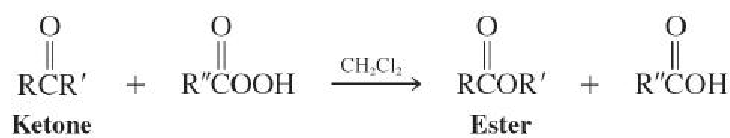
14. Cyanohydrins (Section 17-11)



15. Wittig Reaction (Section 17-12)



16. Baeyer-Villiger Oxidation (Section 17-13)



Migratory aptitudes in Baeyer-Villiger oxidation

Methyl<primary<phenyl~secondary<cyclohexyl<tertiary

Methyl $<$ primary $<$ phenyl ~ secondary $<$ cyclohexyl $<$ tertiary

Important Concepts

1. The **carbonyl group** is the functional group of the **aldehydes (alkanals)** and **ketones (alkanones).** It has precedence over the hydroxy, alkenyl, and alkynyl groups in the naming of molecules.

2. The carbon-oxygen double bond and its two attached nuclei in aldehydes and ketones form a plane. The C═O Câ O unit is **polarized,** with a partial negative charge on oxygen and a partial positive charge on carbon.

3. The 1H NMR $^{1}\mathrm{H}$ NMR **spectra** of aldehydes exhibit a peak at δ≈9.8 ppm. $\delta \approx 9.8$ ppm. In 13C NMR, $^{13}\mathrm{C}$ NMR, the carbonyl carbon absorbs at δ~200 ppm. δ ~ 200 ppm. Aldehydes and ketones have strong infrared bands in the region 1690–1750 cm−1; $1690–1750\ \mathrm{cm}^{-1}$; this absorption is due to the stretching of the C═O Câ O bond. Because of the availability of low-energy n→π* $n \to \pi^*$ transitions, the electronic spectra of aldehydes and ketones have relatively long wavelength bands. This class of compounds displays characteristic mass spectral fragmentations around the carbonyl function.

4. The carbon-oxygen double bond is the subject of **ionic additions.** The catalysts for these processes are acids and bases.

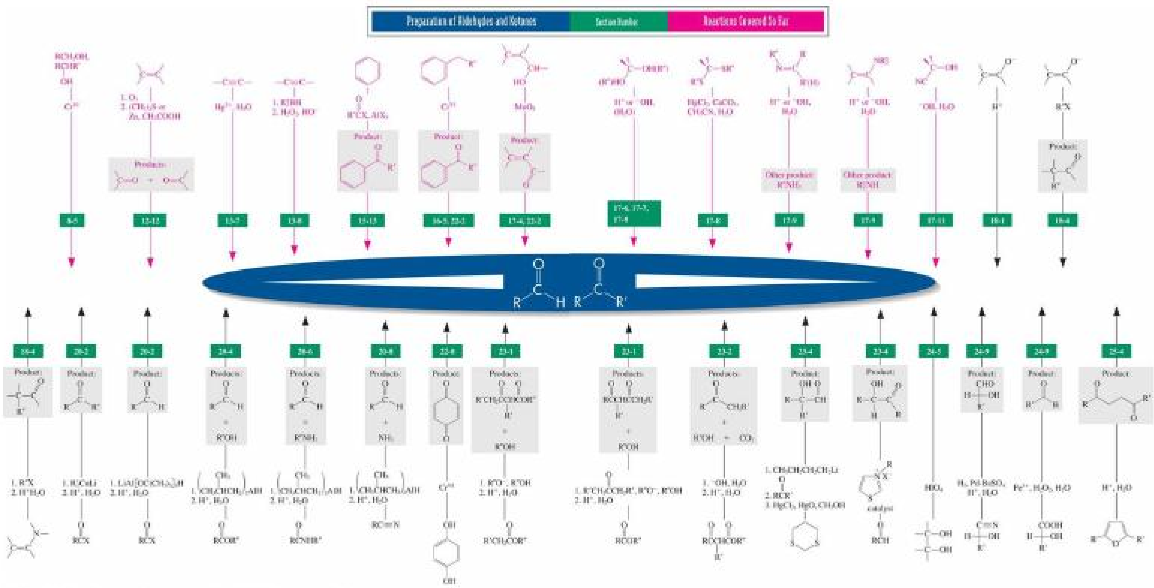
5. The reactivity of the carbonyl group increases with increasing **electrophilic character** of the carbonyl carbon. Therefore, aldehydes are more reactive than ketones.

6. Primary amines undergo **condensation** reactions with aldehydes and ketones to imines; secondary amines condense to enamines.

7. The combination of Friedel-Crafts acylation and Wolff-Kishner or Clemmensen reduction allows synthesis of alkylbenzenes free of the limitations of Friedel-Crafts alkylation.

8. The **Wittig reaction** is an important carbon–carbon bond-forming reaction that produces alkenes directly from aldehydes and ketones.

9. The reaction of **peroxycarboxylic acids** with the carbonyl group of ketones produces **esters.**



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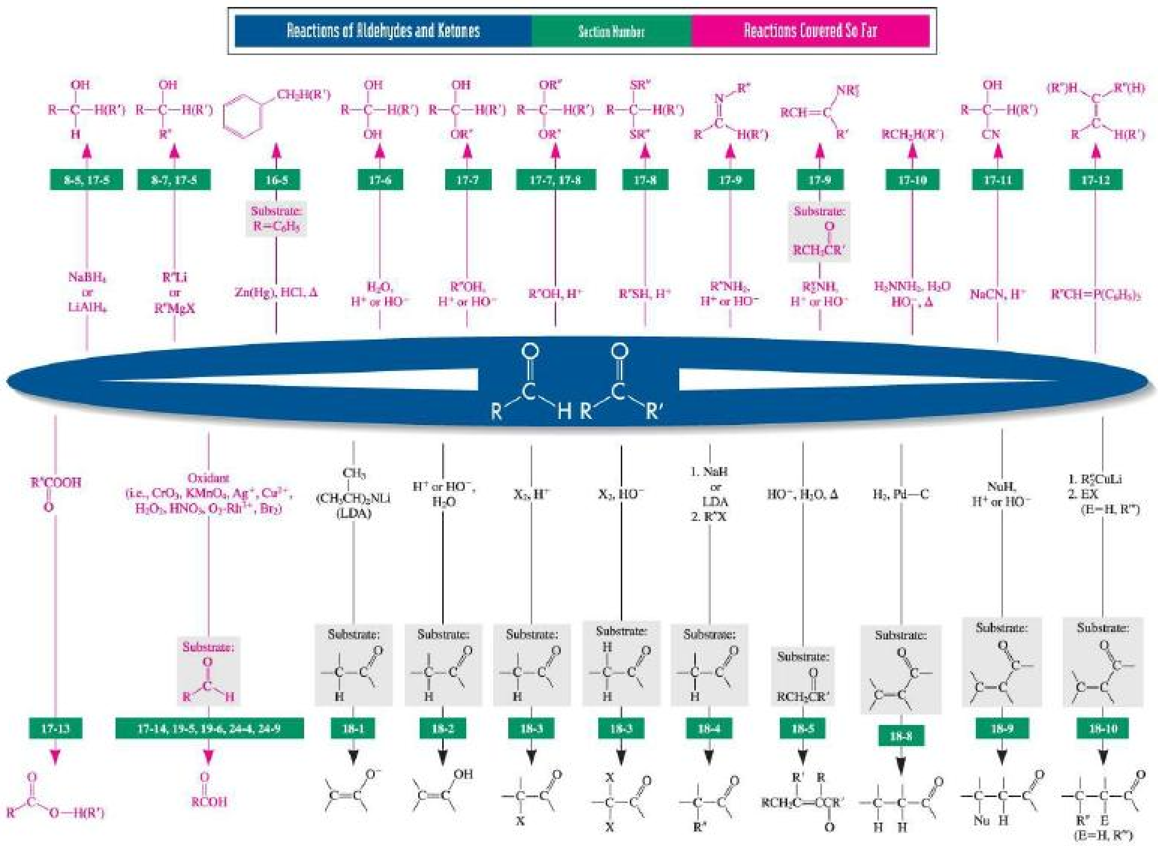

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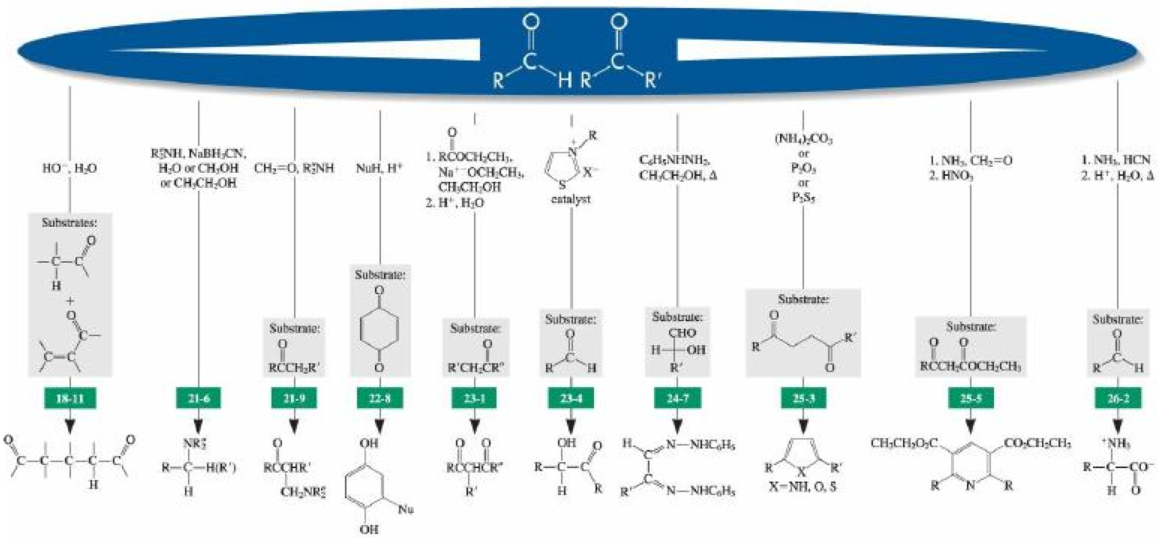
Reactions of Aldehydes and Ketones (continued)	Section Number



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